

Unbounded polynomial norm ratios for super-identical pseudospectra

Abstract

We give explicit nonnegative nilpotent weighted shifts of order $(m+1)^2$ whose singular values agree after every complex scalar shift, but whose norms under a common polynomial have ratio at least $(4/5)\sqrt{m}$. Thus no dimension-independent comparison constant exists. A two-by-two transfer identity proves equality of the shifted singular values; a residue-class decomposition gives the norm bounds. The proof uses only finite matrices and elementary linear algebra. It does not determine the optimal dimension-dependent bound.

1 Statement and construction

All matrix norms are Euclidean operator norms. For $A, B \in \mathbb{C}^{N \times N}$, write $A \sim_{\text{sip}} B$ if

$$s_j(A - zI) = s_j(B - zI) \quad (z \in \mathbb{C}, 1 \leq j \leq N),$$

where singular values are in decreasing order, with multiplicities. This is *super-identical pseudospectra* in the terminology of Fortier Bourque and Ransford [1]. Problem MF-24 [3] asks whether the constants

$$C_N = \sup \left\{ \frac{\|p(A)\|_2}{\|p(B)\|_2} : A, B \in \mathbb{C}^{N \times N}, A \sim_{\text{sip}} B, p \in \mathbb{C}[z], p(B) \neq 0 \right\} \quad (1)$$

are bounded independently of N . Both the matrices and the polynomial may depend on N . Ransford and Walsh [2, Theorem 1.3] prove $C_N \leq \sqrt{N-2}$ for $N \geq 4$. We give a lower bound that diverges.

Theorem 1. *For every integer $m \geq 2$ and real $t > 1$, the weighted shifts $X = X_{m,t}$ and $Y = Y_{m,t}$ defined below have order $N = (m+1)^2$ and satisfy $X \sim_{\text{sip}} Y$. For*

$$D = m + 2, \quad p_m(z) = \sum_{j=1}^m z^{Dj}, \quad (2)$$

one has $p_m(Y) \neq 0$ and

$$\|p_m(X)\|_2 \geq t^2 \sqrt{m}, \quad \|p_m(Y)\|_2 \leq t^2 + m - 1. \quad (3)$$

Consequently,

$$\frac{\|p_m(X)\|_2}{\|p_m(Y)\|_2} \geq \frac{\sqrt{m}}{1 + (m-1)/t^2}. \quad (4)$$

Put $k = m+1$, so $D = k+1$ and $N = k^2 = mD+1$. A weighted shift with word $(w_1, \dots, w_{N-1})$ is the matrix

$$W_{i-1,i} = w_i \quad (1 \leq i < N), \quad W_{r,s} = 0 \quad (s \neq r+1).$$

Thus vertices are numbered $0, \dots, N-1$, and edges $1, \dots, N-1$. Let $U = (t, 1, \dots, 1)$ have length m . Define

$$\begin{aligned} \text{word}(X) &= (U, 1, U, (t^{-1}, U)^{m-1}), \\ \text{word}(Y) &= (U, (t^{-1}, U)^{m-1}, 1, U). \end{aligned} \quad (5)$$

Commas mean concatenation, and the exponent means repetition of the whole parenthesized word. There are $m+1$ motifs U , separated by m bridges: the first bridge of X and the last bridge of Y are 1; all other bridges are t^{-1} . Each word has $(m+1)m + m = N-1$ entries. For example, $m=2$ gives

$$(t, 1, 1, t, 1, t^{-1}, t, 1), \quad (t, 1, t^{-1}, t, 1, 1, t, 1).$$

Both shifts are nonnegative and nilpotent. Also, $\deg p_m = mD = N-1$.

2 Equality of the shifted singular values

Fix a weighted shift W with positive real weights and $z \in \mathbb{C}$. Set $\rho = |z|^2$ and $u = \rho + \eta$, where η is a polynomial variable. The matrix $(W - zI)^*(W - zI) + \eta I$ is tridiagonal. Its diagonal is $u, u + w_1^2, \dots, u + w_{N-1}^2$, and its adjacent off-diagonal entries are $-\bar{z}w_j$ and $-zw_j$. Its leading principal determinants satisfy

$$d_0 = 1, \quad d_1 = u, \quad d_{j+1} = (u + w_j^2)d_j - \rho w_j^2 d_{j-1} \quad (1 \leq j < N). \quad (6)$$

This follows by expanding a tridiagonal determinant along its last row. Define

$$K(a) = \begin{pmatrix} u+a & -\rho a \\ 1 & 0 \end{pmatrix}, \quad v_0 = \begin{pmatrix} u \\ 1 \end{pmatrix}, \quad f = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ -\rho \end{pmatrix}. \quad (7)$$

Then

$$d_N = f^\top K(w_{N-1}^2) \cdots K(w_1^2) v_0. \quad (8)$$

In particular, later edges multiply on the left.

Lemma 2 (Bridge identity). *Let P be a two-by-two matrix over a commutative ring, and let u, ρ, a, b belong to that ring. With the notation in (7), set*

$$R = PK(a), \quad S = PK(b), \quad v = Pv_0.$$

For every integer $n \geq 0$,

$$f^\top R^n S v = f^\top S R^n v. \quad (9)$$

Proof. The transposes are algebraic, not conjugate transposes. Write $w = Pf$. Since $K(a) = v_0 f^\top + a f g^\top$, we have

$$R = v f^\top + a w g^\top, \quad S = v f^\top + b w g^\top.$$

Multiplying these rank-one expressions gives

$$RS - SR = (b - a)[v(f^\top w)g^\top - w(g^\top v)f^\top].$$

After multiplication by $f^\top$ on the left and v on the right, the two scalar terms coincide. Hence $f^\top(RS - SR)v = 0$. The two-by-two Cayley–Hamilton identity

$$R^2 = (\text{tr } R)R - (\det R)I$$

implies by induction that $R^n = \alpha_n R + \beta_n I$ for some ring elements α_n, β_n ; the initial cases are $R^0 = I$ and $R^1 = R$. Substitution proves (9). No division or invertibility is needed. $\square$

The motif U has transfer product $P = K(1)^{m-1}K(t^2)$. Apply the lemma in $\mathbb{C}[\eta]$ with $a = t^{-2}$, $b = 1$, and $n = m - 1$. Reading (5) in the order of (8) gives

$$d_N^X = f^\top R^{m-1} S v = f^\top S R^{m-1} v = d_N^Y.$$

Therefore, for every fixed $z \in \mathbb{C}$,

$$\det((X - zI)^*(X - zI) + \eta I) = \det((Y - zI)^*(Y - zI) + \eta I) \quad (10)$$

as polynomials in η . The two Gram matrices have the same eigenvalues with multiplicities, since their characteristic polynomials agree. They are Hermitian positive semidefinite; taking nonnegative square roots of their eigenvalues proves $X \sim_{\text{sip}} Y$.

3 Polynomial norm bounds

For either word, define the prefix height h by

$$h(0) = 0, \quad w_i = t^{h(i)-h(i-1)}.$$

Every vertex has a unique representation $r = qk + s$, with $0 \leq q, s \leq m$. The two words give

$$h_X(qk + s) = \mathbf{1}_{\{s \geq 1\}} + \mathbf{1}_{\{q \geq 1\}}, \quad h_Y(qk + s) = \mathbf{1}_{\{s \geq 1\}} + \mathbf{1}_{\{q=m\}}. \quad (11)$$

Indeed, the first edge of each motif raises the height by one, its other edges leave it unchanged, and each t^{-1} bridge lowers it by one.

There is a unique path contributing to each potentially nonzero entry of a power of a weighted shift. For an integer $\ell \geq 0$ and $r + \ell < N$,

$$(W^\ell)_{r,r+\ell} = \prod_{i=r+1}^{r+\ell} w_i = t^{h(r+\ell)-h(r)}; \quad (12)$$

all other entries are zero. The monomials of p_m have distinct degrees, so they occupy distinct superdiagonals.

The numerator and a nonzero denominator. For $1 \leq j \leq m$, the vertex $jD = jk + j$ has X -height 2. Thus the first row of $p_m(X)$ contains m entries equal to t^2 , at columns $D, 2D, \dots, mD$. Its Euclidean norm is $t^2\sqrt{m}$, which is at most the operator norm of $p_m(X)$. Also, $h_Y(N-1) = 2$ and $h_Y(0) = 0$, so

$$(p_m(Y))_{0,N-1} = t^2 \neq 0. \quad (13)$$

Residue blocks of the denominator. Every nonzero entry of $p_m(Y)$ joins vertices in the same residue class modulo D . Permuting rows and columns by these classes gives a block diagonal matrix. Since $N = mD + 1$, each block has size $L = m$ or $m + 1$. If the heights along one class, in increasing vertex order, are $a_0, \dots, a_{L-1}$, then (12) gives

$$B_{ij} = \begin{cases} t^{a_j - a_i}, & j > i, \\ 0, & j \leq i. \end{cases} \quad (14)$$

Here every $j > i$ occurs: its vertex separation is $(j-i)D$, with $1 \leq j-i \leq m$, and hence is one of the powers in p_m .

Each Y -block contains at most one height 0 and at most one height 2. To see this, the height-0 vertices are qk for $0 \leq q < m$, with distinct residues $-q$ modulo D . The height-2 vertices are $mk + s$ for $1 \leq s \leq m$, with distinct residues $s - m$ modulo D . All other heights are 1.

The first column and last row of B are zero. Removing them leaves an $(L-1)$ -by- $(L-1)$ matrix $\hat{B}$ with the same operator norm. Explicitly, B maps $(x_0, x_1, \dots, x_{L-1})$ to $(\hat{B}(x_1, \dots, x_{L-1}), 0)$; the coordinate projection and inclusion do not change the supremum defining the norm. Set

$$u_i = t^{-a_i} \quad (0 \leq i \leq L-2), \quad v_j = t^{a_{j+1}} \quad (0 \leq j \leq L-2).$$

Equation (14) implies $0 \leq \hat{B}_{ij} \leq u_i v_j$. For any complex vector x , componentwise absolute values therefore give

$$|\hat{B}x| \leq u(v^\top |x|), \quad \|\hat{B}x\|_2 \leq \|u\|_2 \|v\|_2 \|x\|_2.$$

The second inequality is Cauchy–Schwarz. The restrictions on heights, together with $t > 1$, imply

$$\|u\|_2^2 \leq 1 + (L-2)t^{-2}, \quad \|v\|_2^2 \leq t^4 + (L-2)t^2.$$

Consequently,

$$\|B\|_2 \leq \|u\|_2 \|v\|_2 \leq \sqrt{(1 + (L-2)t^{-2})(t^4 + (L-2)t^2)} = t^2 + L - 2 \leq t^2 + m - 1.$$

The operator norm of a block diagonal matrix is the maximum of the block norms. This proves the second inequality in (3), and completes the proof of Theorem 1.

4 Consequences for the comparison constants

Taking the finite value $t = m$ in (4), and using $4(m - 1) \leq m^2$, yields

$$\frac{\|p_m(X_{m,m})\|_2}{\|p_m(Y_{m,m})\|_2} \geq \frac{4}{5}\sqrt{m} \longrightarrow \infty. \quad (15)$$

Each pair consists of finite rational matrices. Thus the uniform boundedness question has a negative answer, without taking a limit of matrices.

Corollary 3. *For every integer $N \geq 9$,*

$$C_N \geq \sqrt{\lfloor \sqrt{N} \rfloor - 1}. \quad (16)$$

In particular, $\liminf_{N \rightarrow \infty} C_N/N^{1/4} \geq 1$.

Proof. For fixed m , let $t \rightarrow \infty$ in the scalar lower bound in (4). Since $C_{(m+1)^2}$ is a supremum over every finite $t > 1$, it is at least $\sqrt{m}$; attainment is not asserted. Given $N \geq 9$, put $m = \lfloor \sqrt{N} \rfloor - 1 \geq 2$ and pad both $(m+1)^2$ -by- $(m+1)^2$ matrices by the same zero block to reach order N . For every shift z , this adds the same singular values $|z|$ to both lists. Also $p_m(0) = 0$, so the polynomial matrices receive only zero blocks and their norms do not change. This proves (16). The final assertion follows because $(\lfloor \sqrt{N} \rfloor - 1)/\sqrt{N} \rightarrow 1$. $\square$

The exact constants C_N and their optimal growth rate are not determined here. In particular, a gap remains between the lower scale $N^{1/4}$ in Corollary 3 and the upper scale $N^{1/2}$ from [2, Theorem 1.3].

References

- [1] M. Fortier Bourque and T. Ransford, *Super-identical pseudospectra*, J. London Math. Soc. (2) **79** (2009), 511–528. doi:10.1112/jlms/jdn085.
- [2] T. Ransford and N. Walsh, *A four-mean theorem and its application to pseudospectra*, Positivity **26** (2022), article 70. doi:10.1007/s11117-022-00928-8. arXiv:2109.14472v2, 9 July 2022, Theorem 1.3 and Proposition 5.1.
- [3] A. Townsend, *Open Problems in Numerical Linear Algebra*, entry MF-24, accessed 14 September 2026. The entry identifies uniform boundedness as the target and retains the sharp constants as a related quantitative question.